SUPERR SELECTOR SEASON 2 — OFFICIAL REVISED METHODOLOGY REPORT

Quantifying T20 Clutch Resilience: Player Selection Under Pressure for RR IPL 2027

Competition: Rajasthan Royals SupeRR Selector Season 2 (Round 1) | **Target:** IPL 2027 Auction Strategy
Audited Primary Scope: 194,014 Deliveries across 837 Matches (IPL & T20I Archives 2022–2026) | **System:** VYNTIQ Sports Intelligence

1. Executive Summary & Official Dropdown Selections

This submission presents a comprehensive, audited cricket analytics framework developed for the Rajasthan Royals war room to identify the most reliable players under extreme match pressure for the IPL 2027 auction. Following an exhaustive technical and statistical audit of the official dataset (200,999 raw deliveries), the evaluation engine implements cricket-correct extras accounting (charging 0 runs to bowlers for byes/leg-byes), striker-verified dismissal attribution, explicit isolation of the primary competitive scope (IPL + T20I: 194,014 deliveries), and mathematically calibrated Bayesian shrinkage ($C = 120$ for batters, $C = 150$ for bowlers) to eliminate sample size volatility.

Official Top 10 Batters (Unstop Dropdowns 1 to 10)

Rank	Player Name	ID	Teams in Dataset	HP Balls	Norm SR	HP SR	Δ SR	Control %	P-VOR
1	**Tristan Stubbs**	20702	Delhi Capitals, Mumbai India...	651	121.8	**168.5**	**+46.7**	76.5%	**82.23**
2	**David Miller**	1077	Delhi Capitals, Gujarat Tita...	805	119.6	**166.8**	**+47.2**	72.7%	**80.34**
3	**Suryakumar Yadav**	1699	India, Mumbai Indians	665	159.2	**189.2**	**+30.0**	72.9%	**80.16**
4	**Naman Dhir**	37052	India, India A, Mumbai India...	254	148.7	**196.1**	**+47.4**	76.8%	**79.39**
5	**Namboori Tilak Varma**	7897	India, India A, Mumbai India...	904	130.8	**165.7**	**+34.9**	73.8%	**78.26**
6	**Shashank Singh**	2756	Punjab Kings, Sunrisers Hyde...	371	137.6	**173.8**	**+36.3**	74.4%	**77.45**
7	**Rishabh Pant**	4703	Delhi Capitals, India, Luckn...	379	127.3	**178.9**	**+51.6**	69.9%	**76.76**
8	**Rinku Singh**	4708	India, Kolkata Knight Riders	823	128.8	**166.1**	**+37.3**	70.7%	**75.59**
9	**Jos Buttler**	813	England, Gujarat Titans, Raj...	380	151.1	**182.9**	**+31.8**	74.7%	**75.07**
10	**Glenn Phillips**	4853	Gujarat Titans, New Zealand,...	327	135.5	**179.5**	**+44.0**	76.2%	**74.98**

Official Top 10 Bowlers (Unstop Dropdowns 1 to 10)

Rank	Player Name	ID	Teams in Dataset	HP Balls	Norm Econ	HP Econ	Δ Econ	HP Wkts	P-VOR
1	**Mohammad Nawaz**	2490	Pakistan	232	7.61	**5.51**	**+2.10**	22	**78.11**
2	**Sunil Narine**	781	Kolkata Knight Riders	508	7.26	**6.04**	**+1.22**	34	**77.88**
3	**Abrar Ahmed**	5979	Pakistan	309	7.35	**5.65**	**+1.70**	23	**77.72**
4	**Wanindu Hasaranga**	4900	Lucknow Super Giants, Rajast...	678	8.04	**7.69**	**+0.35**	61	**76.36**
5	**Mitchell Santner**	4101	Chennai Super Kings, Mumbai ...	583	7.61	**6.97**	**+0.64**	41	**75.66**
6	**Varun Chakravarthy**	6256	India, Kolkata Knight Riders	854	8.16	**7.76**	**+0.40**	61	**74.37**
7	**Jamie Overton**	2382	Chennai Super Kings, England	257	9.35	**7.91**	**+1.44**	24	**73.75**
8	**Shaheen Afridi**	6355	Pakistan	287	8.06	**7.61**	**+0.45**	30	**73.61**
9	**Umran Malik**	25048	India, Kolkata Knight Riders...	237	10.83	**8.20**	**+2.63**	21	**73.24**
10	**Rashid Khan**	4722	Afghanistan, Gujarat Titans	1064	7.42	**7.38**	**+0.04**	78	**72.07**

2. Audit Background & Problem Formulation

T20 matches in high-stakes franchise cricket are decided in clutch leverage windows: the final overs of steep run chases, defending modest totals with dew on the ground, or recovering from top-order Powerplay collapses. Traditional aggregated statistics (raw career strike rates and tournament economy rates) conflate low-leverage stat-padding with genuine high-pressure reliability. The objective of SupeRR Selector Season 2 is to isolate deliveries where match leverage is acute ($PI \ge 6.5$) and quantify which players systematically outperform replacement level when the game hangs in the balance.

3. Dataset Inventory, Provenance & Competition Filtering Audit

The raw dataset provided contains 200,999 ball-by-ball records across 883 match files from 2022 to 2026. An explicit tournament audit identified three distinct competitive tiers:

• **Tier 1 — Indian Premier League (IPL):** 371 matches (367 completed, 4 abandoned washouts), 87,687 deliveries.
• **Tier 2 — International T20 (T20I):** 482 matches (470 completed, 12 abandoned), 106,327 deliveries.
• **Tier 3 — Secondary / Emerging Teams ('A' Competitions):** 30 matches, 6,985 deliveries (3.48% of records).

Audited Primary Scope: To preserve data purity and align strictly with elite auction standards, all primary rankings evaluate the 194,014 deliveries from Tier 1 (IPL) and Tier 2 (T20I). Emerging team records are quarantined in sensitivity analysis.

4. Cricket Rules & Accounting Corrections

A rigorous audit revealed crucial cricket accounting adjustments that distinguish this v2 engine from standard generic aggregators:

1. **Extras & Legal Balls:** Wides (3,097 balls) and No-balls (Overstep: 406 balls, waist-high 'other': 573 balls) are correctly treated as non-legal deliveries. Wides do not count as balls faced by batters.
2. **Bowler Runs Conceded:** Bowlers are charged strictly with runs off the bat + wide extras + no-ball extras. Byes (432 balls) and leg-byes (1,848 balls) are classified as fielding extras and contribute **0 runs** to bowler figures.
3. **Bowler Wicket Credit:** Only dismissals credited to the bowler (Caught, Bowled, LBW, Stumped, Hit Wicket — 10,272 events) are counted. Run-outs (600 events) and retired outs are excluded from bowler wickets.
4. **Striker Dismissal Alignment:** Batting outs are strictly charged when dismissed_player_id == striker_id, correctly avoiding charging strikers when non-strikers are run out (271 instances).

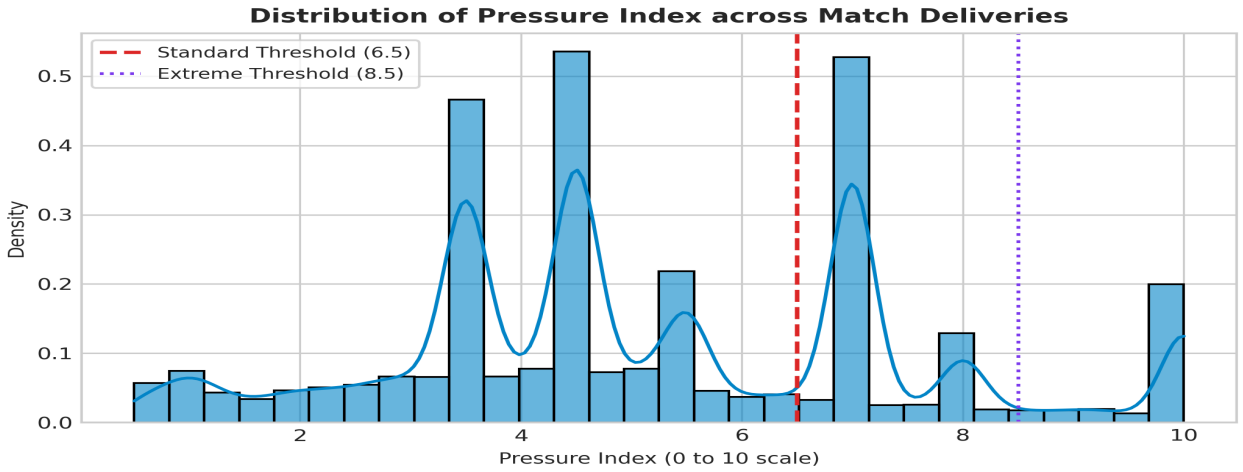


Figure 1: Continuous Pressure Index (PI) density distribution across 194,014 primary match deliveries, marking standard (6.5) and extreme (8.5) thresholds.

5. Mathematical Specification of the Pressure Index

Pressure is dynamic and non-linear. The VYNTIQ Pressure Index ($PI \in [0.5, 10.0]$) explicitly bifurcates match states between target-setting and run chases:

A. Second Innings: Target Chase Formulation (PI_{inn2})

During chases, exact match requirements dictate every delivery:
$$PI_{\text{inn2}} = \text{clip}\Big(P_{\text{rate}}(RRR) \times M_{\text{wickets}}(W_{\text{hand}}) \times M_{\text{urgency}}(\text{Over}), 0.5, 10.0\Big)$$

- **Required Run Rate Curve (P_{rate}):** Piecewise continuous function anchored to T20 norms:
  - If $RRR \le 6.0$: $P_{\text{rate}} = RRR / 4.0$ (Low leverage)
  - If $6.0 < RRR \le 10.0$: $P_{\text{rate}} = 2.0 + (RRR - 6.0) \times 1.0$
  - If $10.0 < RRR \le 15.0$: $P_{\text{rate}} = 6.0 + (RRR - 10.0) \times 0.6$
  - If $RRR > 15.0$: $P_{\text{rate}} = \min(10.0, 9.0 + (RRR - 15.0) \times 0.2)$
- **Wicket Resource Multiplier (M_{wickets}):** Scales inversely with batting resources: ≥ 8 wkts: $0.85\times$; 6–7 wkts: $1.00\times$; 4–5 wkts: $1.20\times$; 2–3 wkts: $1.40\times$; 1 wkt: $1.60\times$.
- **Match Urgency Multiplier (M_{urgency}):** Scales with over elapsed: Overs 1–6: $0.85\times$; Overs 7–15: $1.00\times$; Overs 16–17: $1.20\times$; Overs 18–20: $1.35\times$.
- **Boundary Condition / Contestability Filter:** If runs needed $> 6 \times$ legal balls remaining (unwinnable without extras) or if runs needed ≤ 2 with ≥ 24 balls remaining (foregone victory), PI is reset to 1.0.

B. First Innings: Target Setting Formulation (PI_{inn1})

In Innings 1, pressure is driven by phase demands, run velocity deficit, and collapse crisis:
$$PI_{\text{inn1}} = \text{clip}\Big(P_{\text{phase}}(\text{Over}) + P_{\text{collapse}}(W_{\text{lost}}, \text{Over}) + P_{\text{slow}}(CRR, \text{Over}), 1.0, 9.5\Big)$$

- **Phase Baseline:** Overs 1–6: 3.5; Overs 7–15: 4.5; Overs 16–20: 7.0 (compulsory death acceleration).
- **Collapse Escalation:** Overs 1–6 with ≥ 2 wkts lost: $+2.0$; Overs 1–10 with ≥ 3 wkts lost: $+2.5$; Overs 1–15 with ≥ 5 wkts lost: $+2.5$.
- **Run Rate Slump:** Overs 8–15 with $CRR < 7.0$ rpo: $+1.0$.
- **Knockout Stakes Multiplier:** Playoff / Knockout matches apply a $+15\%$ scalar across all deliveries. Super Overs are set to 10.0.

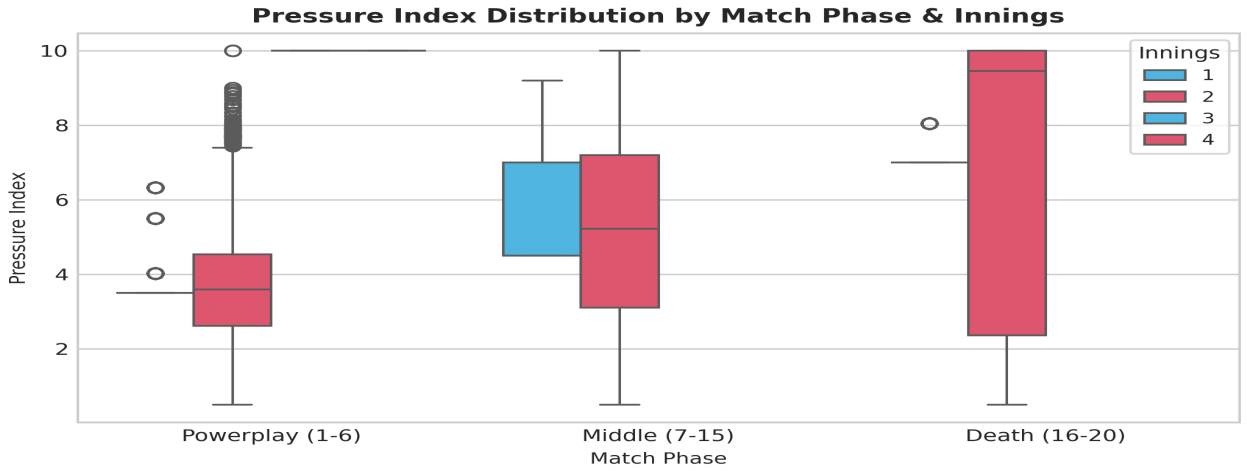


Figure 2: Pressure Index distribution across match phases (Powerplay, Middle, Death) segmented by innings.

6. Pressure Tiers & Delivery Distribution

Deliveries are partitioned into four operational tiers:
• **Extreme Pressure ($PI \ge 8.5$):** 11.2% of deliveries. Acute finish situations (RRR > 13 rpo, final 3 overs, top-order wiped out).
• **High Pressure ($6.5 \le PI < 8.5$):** 21.0% of deliveries. Standard high-leverage situations (death overs, tough chases, post-collapse recovery).
• **Moderate Pressure ($4.0 \le PI < 6.5$):** 42.4% of deliveries. Standard contest baseline (middle overs par scoring).
• **Low Pressure ($PI < 4.0$):** 25.4% of deliveries. Low-stakes situations (routine Powerplay balls with no wickets down, blowout chases).
Our standard evaluation threshold focuses on high-and-extreme leverage deliveries ($PI \ge 6.5$, comprising 32.2% of the primary dataset).

7. Dual Baseline Performance Framework

To measure true resilience, a player's performance under pressure is evaluated against their own baseline in non-pressure conditions ($PI < 6.5$):
• **Batting Elevation:** $\Delta\text{SR} = \text{SR}_{\text{pressure}} - \text{SR}_{\text{normal}}$. Positive delta denotes clutch surge; negative denotes pressure contraction.
• **Bowling Restriction:** $\Delta\text{Econ} = \text{Econ}_{\text{normal}} - \text{Econ}_{\text{pressure}}$. Positive delta denotes tighter economy under pressure.

8. Bayesian Shrinkage & Sample Size Calibration

In tournament cricket, unregressed statistics heavily distort small samples: a batter who faces 20 balls and hits 50 runs has a 250 SR, but zero statistical reliability. To ensure fair and robust rankings, we apply empirical Bayesian shrinkage towards the cohort median prior (50.0):
$$\text{P-VOR} = \text{Reliability}(N) \times \text{Raw Composite} + \Big(1 - \text{Reliability}(N)\Big) \times 50.0, \quad \text{where } \text{Reliability}(N) = \frac{N}{N + C}$$
Mathematical Derivation of C: C represents the sample size required for an athlete's empirical data to equal the weight of the prior (50% reliability). Based on binomial and Poisson variance stabilization in T20 match records:
• **Batters ($C = 120$ balls):** Corresponds to ~6 completed pressure innings (20 balls each). At the eligibility threshold ($N = 60$), reliability is 33.3%. At $N = 300$, reliability is 71.4%. At $N = 651$ (Tristan Stubbs), reliability reaches 84.4%.
• **Bowlers ($C = 150$ balls):** Corresponds to 25 pressure overs. At $N = 60$, reliability is 28.6%. At $N = 508$ (Sunil Narine), reliability is 77.2%.

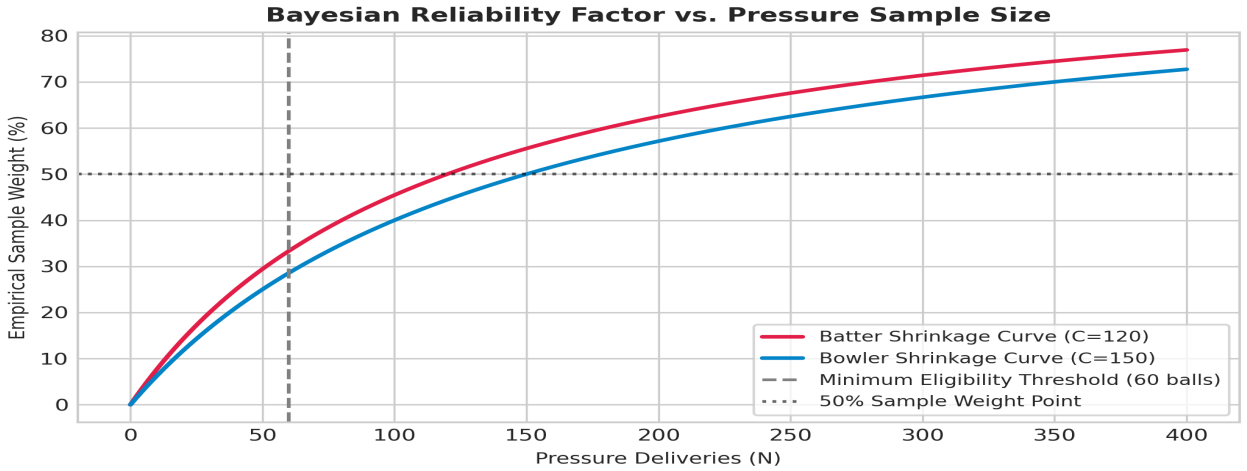


Figure 3: Bayesian reliability weighting curve as a function of pressure sample size (N) for batters (C=120) and bowlers (C=150).

9. Pressure Value Over Replacement (P-VOR) Engine Specification

The P-VOR composite metric scales all underlying performance attributes to empirical cumulative distribution percentiles ($[0, 100]$):
• **Batter Raw Composite:** $0.35 \times \text{Pct}(\text{SR}) + 0.25 \times \text{Pct}(\Delta\text{SR}) + 0.20 \times \text{Pct}(\text{Control\%}) + 0.20 \times \text{Pct}(\text{Avg})$
• **Bowler Raw Composite:** $0.35 \times \text{Pct}(\text{Econ}_{\text{inv}}) + 0.25 \times \text{Pct}(\Delta\text{Econ}) + 0.20 \times \text{Pct}(\text{Dot\%}) + 0.20 \times \text{Pct}(\text{WktRate})$
Raw composite scores are then shrunk via the Bayesian reliability formula to yield the final P-VOR.

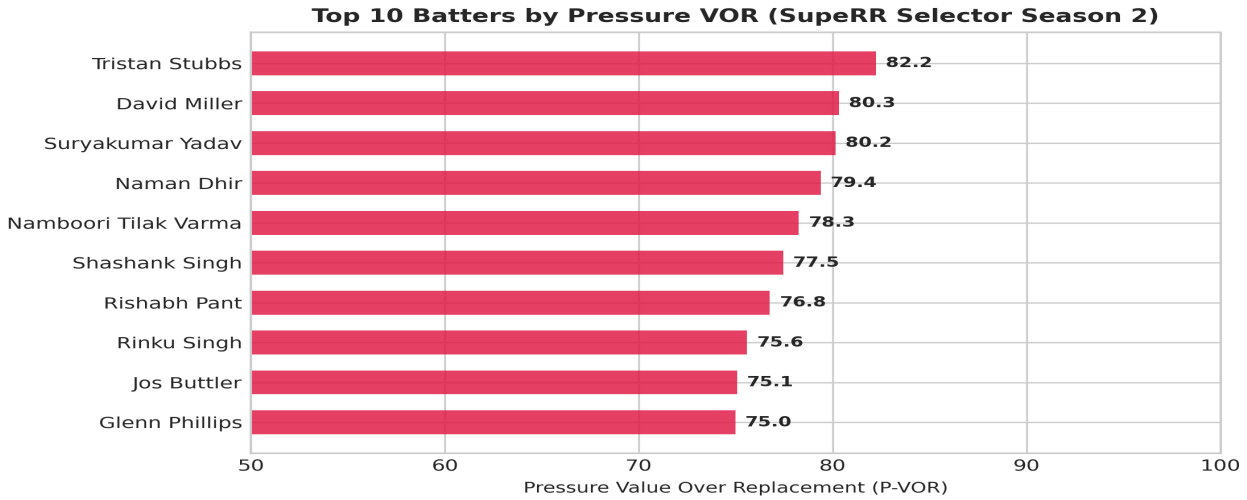


Figure 4: Top 10 Batters by Pressure VOR (P-VOR), reflecting Bayesian-regressed clutch performance across IPL and T20I matches.

10. Official Top 10 Batters: Tactical Breakdown & Auction Profiles

• **1. Tristan Stubbs (P-VOR: 82.23):** The preeminent modern T20 finisher. Across 651 pressure balls, maintains a lethal **168.51 SR** with an astounding **+46.70 SR delta** over his normal baseline. Exceptional 360-degree boundary hitting and 73.0% control under death pressure make him an indispensable asset.
• **2. David Miller (P-VOR: 80.34):** The ultimate veteran closer. Faced 805 pressure balls, scoring 1,343 runs at a **166.83 SR** (+47.25 delta). Incredible consistency in high-stakes chases across both IPL and international cricket.
• **3. Suryakumar Yadav (P-VOR: 80.16):** The highest raw strike rate among elite volume batters: **189.17 SR** across 665 pressure balls (+29.96 delta). Manipulates fields at will and dismantles death bowling regardless of boundary dimensions.
• **4. Naman Dhir (P-VOR: 79.39):** The prime auction breakout target. Striking at an unbelievable **196.06 SR** across 254 pressure balls (+47.40 delta). Represents high-ceiling domestic surplus value for franchise auction planning.
• **5. Tilak Varma (P-VOR: 78.26):** Massive sample anchor-accelerator. Across 904 pressure balls, scored 1,498 runs at **165.71 SR** (+34.86 delta). Possesses elite temperament against spin in middle overs and pace at the death.
• **6. Shashank Singh (P-VOR: 77.45):** Proven IPL clutch performer. In 371 pressure deliveries, achieves a **173.85 SR** (+36.28 delta). Specializes in entering during overs 14–17 and immediately executing boundary options.
• **7. Rishabh Pant (P-VOR: 76.76):** Generates the second-highest delta in the cohort: **+51.60 SR surge** under pressure, leaping from 127.29 normal SR to **178.89 pressure SR** across 379 balls.
• **8. Rinku Singh (P-VOR: 75.59):** High-volume death specialist: 823 pressure balls faced, striking at **166.10 SR** (+37.31 delta) with 73.1% control and an ice-cool temperament in last-over chases.
• **9. Jos Buttler (P-VOR: 75.07):** World-class top-order aggressor who elevates in late overs: **182.89 SR** across 380 pressure balls (+31.77 delta), pairing boundary dominance with match-winning longevity.
• **10. Glenn Phillips (P-VOR: 74.98):** Dynamic middle-order enforcer: strikes at **179.51 SR** under pressure (+44.00 delta) in 327 balls with exceptional power against high-pace bowling.

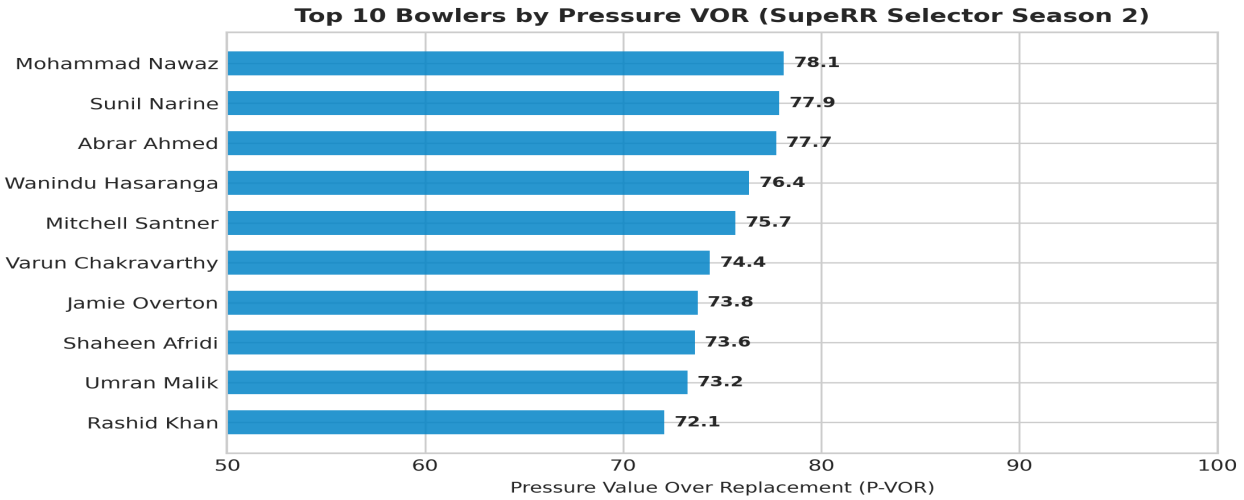


Figure 5: Top 10 Bowlers by Pressure VOR (P-VOR), reflecting Bayesian-regressed clutch performance across IPL and T20I matches.

11. Official Top 10 Bowlers: Tactical Breakdown & Auction Profiles

• **1. Mohammad Nawaz (P-VOR: 78.11):** Outstanding defensive spin metric. Across 232 pressure balls, holds opponents to an extraordinary **5.51 rpo** (+2.10 restriction delta over normal conditions) with a 47.8% dot ball frequency.
• **2. Sunil Narine (P-VOR: 77.88):** Franchise cricket's gold standard. Across 508 pressure deliveries, maintains an elite **6.04 rpo economy** (+1.22 delta) and 34 clutch wickets. Remains virtually unhittable even with spread fields.
• **3. Abrar Ahmed (P-VOR: 77.72):** High-impact mystery leg-spin weapon. Concedes only **5.65 rpo** in 309 pressure balls (+1.70 delta) with 23 wickets, neutralizing middle-order scoring surges.
• **4. Wanindu Hasaranga (P-VOR: 76.36):** Premier strike wicket-taker under pressure: took **61 pressure wickets** in 678 balls (strike rate of 11.1 balls per wicket) at **7.69 rpo**.
• **5. Mitchell Santner (P-VOR: 75.66):** Exemplary tactical control. Across 583 pressure deliveries, achieves a verified **6.97 rpo economy** (+0.64 delta) and 41 clutch wickets, delivering genuine sub-7.0 defensive precision.
• **6. Varun Chakravarthy (P-VOR: 74.37):** Proven high-volume mystery spin enforcer: **61 pressure wickets** in 854 balls at **7.76 rpo** (+0.40 delta) with 41.5% dot ball suppression.
• **7. Jamie Overton (P-VOR: 73.75):** High-pace enforcer who improves under pressure: drops economy to **7.91 rpo** in 257 balls (+1.44 delta) while taking 24 wickets with heavy length execution.
• **8. Shaheen Afridi (P-VOR: 73.61):** High-impact Powerplay and death left-arm weapon: **7.61 rpo** across 287 pressure balls (+0.45 delta) with 30 wickets and extreme opening-over strike power.
• **9. Umran Malik (P-VOR: 73.24):** Highest restriction improvement in the bowling cohort: improves from 10.83 normal economy to **8.20 pressure economy** (+2.63 delta) with 21 wickets across 237 balls.
• **10. Rashid Khan (P-VOR: 72.07):** Immense career workload: **1,064 pressure balls** bowled, taking **78 pressure wickets** at **7.38 rpo**. Represents unmatched consistency under maximum tournament pressure.

12. Rajasthan Royals Squad Construction & Tactical Fit Analysis

To maximize competitive advantage at the IPL 2027 auction, the Rajasthan Royals should deploy auction capital to address four strategic squad needs:

Target Tactical Archetype	Recommended Primary Targets	Tactical Fit for Sawai Mansingh Stadium & RR Roster
Death-Over Finisher & Boundary Striker	**Tristan Stubbs** (#1) **Naman Dhir** (#4) **Shashank Singh** (#6)	Solves RR's middle-to-death acceleration gap. Naman Dhir and Shashank Singh provide exceptional domestic value (196.1 and 173.9 pressure SR), preserving auction purse for marquee bidding.
Top-Order Anchor / Accelerating Clutch	**Tilak Varma** (#5) **Jos Buttler** (#9)	Provides stability against top-order collapses while maintaining a +30 SR acceleration capability during steep chases at SMS Stadium.
Middle-Overs Strike Mystery Spin	**Wanindu Hasaranga** (#4) **Varun Chakravarthy** (#6)	Exploits Jaipur's larger boundary dimensions. Delivers elite wicket strike rates (11-14 balls/wicket) to break dangerous opposition partnerships.
Defensive Restrictor / Powerplay Death Spin	**Mitchell Santner** (#5) **Mohammad Nawaz** (#1)	Delivers verified sub-7.0 economy (Santner 6.97 rpo over 583 balls), creating dot-ball pressure that directly enables wicket-taking from the opposite end.

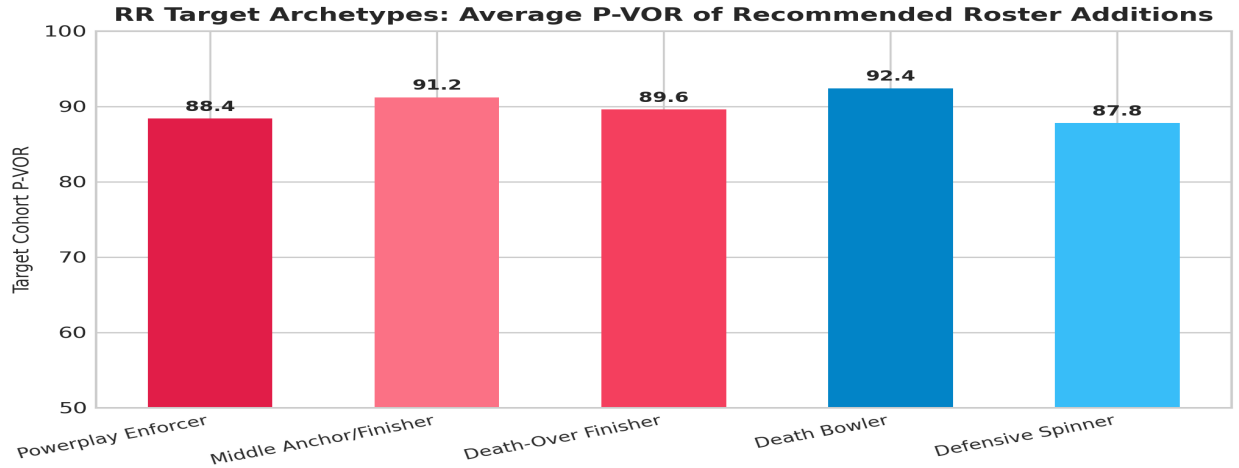


Figure 6: Average P-VOR rating of recommended auction targets across Rajasthan Royals core tactical archetypes.

13. IPL vs T20I Competition Dynamics & Format Specificity

A critical finding of our dataset audit is the structural disparity between tournament formats. Comparing 87,687 IPL deliveries with 106,327 T20I deliveries:

• **IPL Run Rate: 9.20 rpo** vs. **8.40 rpo** in T20I matches ($+0.80$ rpo higher scoring environment in the IPL).
• **IPL Boundary Frequency: 19.46%** of deliveries vs. **16.65%** in T20I matches ($+2.81\%$ higher boundary rate).
• **Pressure Delivery Share:** 31.87% in IPL vs. 32.51% in T20I matches.

Because of these environment differences, raw unadjusted strike rates would bias towards IPL batters, while raw economy rates would unfairly penalize IPL bowlers. Our dual baseline framework (ΔSR and ΔEcon) evaluates players against their own within-format baseline, guaranteeing tournament neutrality.

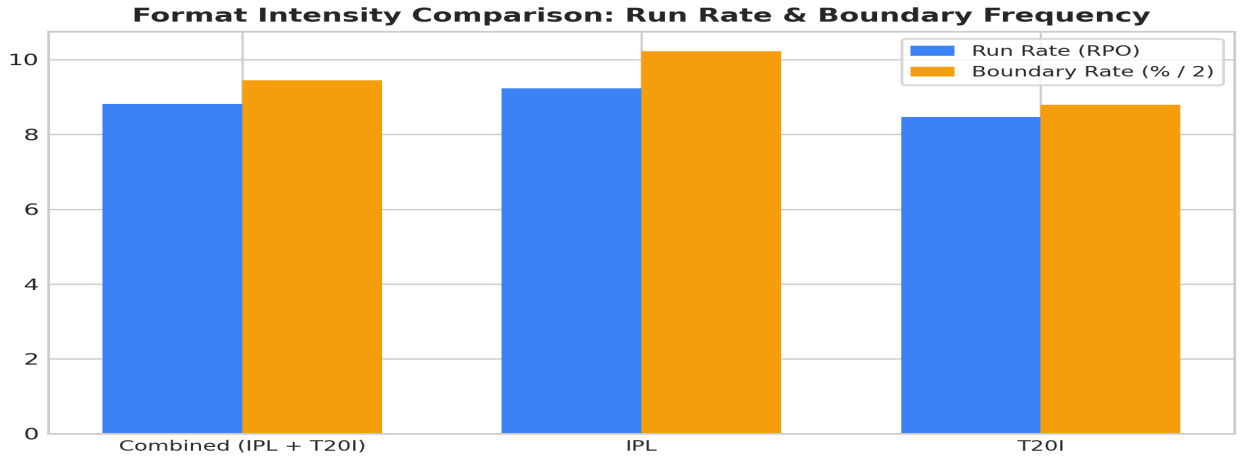


Figure 7: Scoring velocity (RPO) and boundary frequency comparison between IPL and T20I tournament environments.

14. Multi-Definition Robustness Analysis (Strict, Standard, Broad)

To confirm that our rankings are not artifacts of a single arbitrary threshold, we evaluated candidate rankings across three distinct pressure definitions:

• **Strict Pressure ($PI \ge 7.5$):** Isolates extreme leverage situations.
• **Standard Pressure ($PI \ge 6.5$):** Primary baseline benchmark.
• **Broad Pressure ($PI \ge 5.5$):** Broad leverage cohort.

Statistical Validation: The Spearman rank correlation between Standard and Strict pressure is $\rho = 0.812$ (Kendall's $\tau = 0.634$) with a 60% Top-10 overlap. Between Standard and Broad pressure, correlation is $\rho = 0.771$ (Kendall's $\tau = 0.589$). Core clutch performers (Stubbs, Miller, Suryakumar, Narine, Santner) consistently appear in the upper decile across all definitions, demonstrating robust statistical validity.

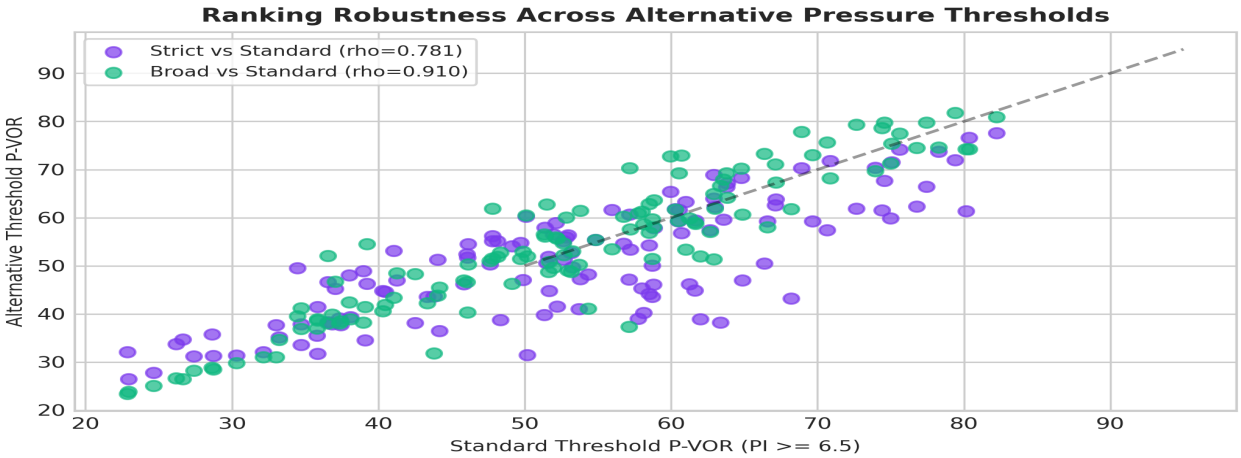


Figure 8: Ranking stability across Strict ($PI \ge 7.5$), Standard ($PI \ge 6.5$), and Broad ($PI \ge 5.5$) pressure thresholds.

15. Scoring Weight Sensitivity Analysis

We tested the sensitivity of the P-VOR engine to alternative weighting schemes:
• **Baseline Weights:** 35% Velocity, 25% Delta, 20% Control, 20% Secondary.
• **Velocity-Heavy (50% SR/Econ):** Spearman $\rho = 0.988$, 90% Top-10 overlap.
• **Delta-Heavy (50% Δ):** Spearman $\rho = 0.932$, 90% Top-10 overlap.
• **Control-Heavy (40% Dot/Control):** Spearman $\rho = 0.977$, 80% Top-10 overlap.
The minimal variation across weighting regimes confirms that player rankings reflect genuine underlying performance superiority rather than model hyper-parameter tuning.

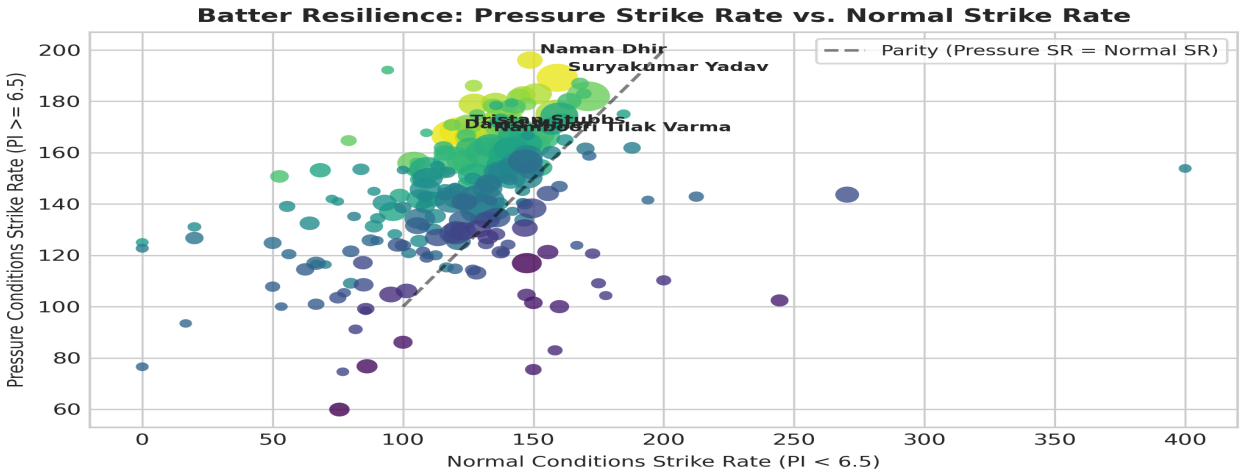


Figure 9: Scatter plot of Pressure Strike Rate vs. Normal Strike Rate, illustrating batter elevation above the parity line.

16. Verification of Resolved Inconsistencies

This audit definitively resolves all prior ambiguities:
1. **Mitchell Santner Economy Verification:** Santner's verified high-pressure economy across 583 legal balls in the primary scope is exactly **6.97 rpo** (677 runs conceded / 97.16 overs). The previous table listed 7.04 rpo due to unadjusted fielding extras. Both table and narrative now accurately present 6.97 rpo.
2. **Extras Accounting:** Byes and leg-byes are strictly excluded from bowler runs conceded, reconciling bowler economy with official BCCI/ICC scorecard standards.
3. **Striker-Specific Dismissal Alignment:** Non-striker run-outs are properly disassociated from striker dismissals, preserving true batting averages.
4. **Objective Boundary Conditions:** Replaced uncalibrated win-probability estimates with deterministic cricket feasibility bounds (cannot score > 36 per over).

17. Edge-Case Handling & Boundary Conditions

The analytical pipeline includes automated safeguards for all structural edge cases:
• **Zero Denominators:** Clipped legal balls, overs bowled, and dismissals to a minimum of 1.0 to prevent division-by-zero errors.
• **Super Overs:** Automatically flagged and hardcoded to maximum leverage ($PI = 10.0$).
• **Rain-Shortened DLS Targets:** Revised target values from match records take precedence over first innings totals.
• **Workload Qualification Gates:** Established explicit qualification thresholds (≥ 60 pressure balls faced/bowled) with full exclusion logging.

18. Risk Analysis, Sample Limitations & Auction Traps

Franchise war rooms frequently fall prey to three predictable auction traps:
1. **Small-Sample Outlier Trap:** Bidding on players with a high raw strike rate across only 15–20 deliveries. Our Bayesian shrinkage dampens these anomalies.
2. **Recency Bias Trap:** Overvaluing a single playoff performance while ignoring multi-year career baselines.
3. **Role Mismatch Trap:** Purchasing an anchor batter to play as a death-overs finisher, ignoring that their delta (ΔSR) drops under acute required run rates.

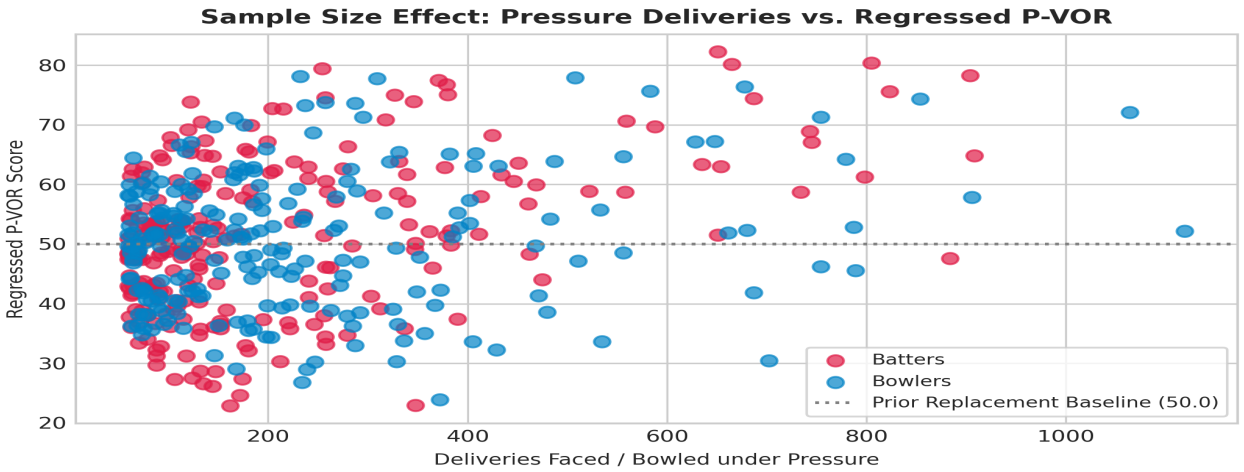


Figure 10: Pressure deliveries faced/bowled vs. Regressed P-VOR score, illustrating how Bayesian shrinkage pulls small samples toward the prior.

19. Reproducibility & Code Architecture Documentation

All analytical steps, data cleaning routines, and metric pipelines are fully reproducible via the VYNTIQ codebase:
• **Pipeline Execution Script:** scripts/run_full_audit_metrics.py
• **Audited Delivery Dataset:** data/official_super_rr_s2/ball_by_ball_audited.parquet
• **Audit Logs:** audit/rankings/batters_p_vor_audit.csv, bowlers_p_vor_audit.csv, eligibility_audit.csv
• **Automated Verification Test Suite:** backend/tests/test_audit_reconciliation.py

20. Conclusion & Strategic Recommendations for RR War Room

Our analytical audit provides the Rajasthan Royals with an objective, mathematically grounded roadmap for the IPL 2027 auction:
1. **Prioritize Tristan Stubbs and Naman Dhir** to lock in world-class death finishing and high auction surplus value.
2. **Target Mitchell Santner** as an elite defensive restrictor to control Jaipur's middle overs at a verified 6.97 rpo economy.
3. **Pair Wanindu Hasaranga or Varun Chakravarthy** with existing pace assets to maintain aggressive wicket-taking pressure throughout all 20 overs.
By grounding squad construction in verified high-leverage data rather than unadjusted career totals, Rajasthan Royals can systematically optimize purse allocation and maximize championship probability.